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संगणक विज्ञान और अभियांत्रिकी विभाग
Department of Computer Science and Engineering
Introduction to Unix and Software Tools (NCSC101)
End Semester Exam , Date: September 17, 2025
B.Tech 1st Year

Timing: 6:00PM - 7:00PM Monsoon Semester 2025-2026 Maximum marks: 30

1. Briefly explain the difference between `>` and `>>`. In one line each, give an example that (i) overwrites a file with command output and (ii) appends to a log file. (3)

Solution:

- `>` redirects standard output and overwrites the target file (**creates** it if absent).
- `>>` redirects standard output and appends to the target file (**creates** it if absent).

Examples:

```
# (i) overwrite:
```

```
ls -l > files.txt
```

```
# (ii) append to a log:
```

```
echo "$(date '+%F %T') - job started" >> app.log
```

2. What do the anchors `^` and `$` mean in regular expressions? Write a single regular expression to match a line containing exactly three digits. Your solution is allowed to use only one class and alteration(`|`) is not allowed. (3)

Solution: `^` matches the start of the line; `$` matches the end of the line. A regex that matches a line with exactly three digits is:

$$^{[0-9]}{3}\$$$

(uses one character class `[0-9]` and anchors).

3. In the awk pattern-action model, what do the special variables `$0`, `$1`, `NF`, and `NR` represent, and how can you write a one-line awk command to print only the last field of every line? (3)

Solution:

`$0`: the entire current record (whole line).

\$1: the first field of the current record.

NF: number of fields in the current record.

NR: current record number (line number across input).

Print the last field of each line:

```
awk '{ print $NF }' input.txt
```

4. What are two fundamental modes in the Vim editor, and what are two common normal-mode motion commands? Additionally, provide the single command required to find every occurrence of the string foo and replace it with the string bar throughout the entire file. (3)

Solution: Modes: Normal mode, Insert mode. **Common motions (normal mode):** w (next word start), \$ (end of line); other valid answers include 0, b, gg, G.

Replace all foo with bar in the whole file:

```
:%s/foo/bar/g
```

5. Consider the file sample.txt with the content shown below: (3+2)

```
1 INFO: System startup sequence initiated.
2 Warning: Disk space is running low.
3 This line contains an Error: but not at the start.
4 Error: Failed to connect to database server.
5 Checking all systems.
6 A Warning without a colon.
7 error: Case-sensitive test, will not match.
8 DEBUG: User 'admin' logged in.
9 Error:
10 Error: Critical process terminated unexpectedly.
```

Write a single grep command that uses extended regular expressions to print the line numbers of all lines that begin with either Error or Warning, followed immediately by a colon. Then, write the exact output produced by your command.

Solution: Command:

```
grep -nE '^(Error|Warning):' sample.txt
```

Exact output:

```
2:Warning: Disk space is running low.
4>Error: Failed to connect to database server.
9>Error:
10>Error: Critical process terminated unexpectedly.
```

6. Write a shell script that accepts **two numbers** as command line arguments and prints their **sum**. The script should **validate** that the user has provided exactly two arguments. If the user does not provide two arguments, the script must display a proper usage message in the format. Also explain how to execute (run) this script from the command line. **(3+3+1)**

Solution: Script (sum.sh):

```
#!/usr/bin/env bash
# sum.sh - print sum of two numbers

if [ "$#" -ne 2 ]; then
    echo "Usage: $0 num1 num2" >&2
    exit 1
fi

# Integer arithmetic; works for integers (and shell supports
# +/-)
sum=$(( $1 + $2 ))
echo "$sum"
```

How to run:

```
chmod +x sum.sh          # make executable (one-time)
./sum.sh 12 30           # prints: 42

# Alternatively (no chmod needed):
bash sum.sh 12 30
```

7. Consider the file report.txt with the content shown below: **(3+3)**

```
1 2025-06-12 server1: error: disk full; error: retry failed;
   error: giving up
2 note: diagnostics complete
3 warning: error error ERROR error
4 error,error;error - still investigating
5 error occurred but error not critical
6 error error error error error
7 Error once; error once
```

Write a single sed command that (i) replaces *only the third occurrence* of the exact, lowercase string error on each line with WARNING, and (ii) prints *only those lines on which the substitution occurs* (suppress all other lines). Use one sed invocation (no in-place editing, no pipes). Then, write the exact output produced by your command.

Solution: Command (GNU sed):

```
sed -n 's/\<error\>/WARNING/3p' report.txt
```

or

```
sed -n 's/error/WARNING/3p' reports.txt
```

Explanation: `\<error\>` matches the exact lowercase word `error`; the `3` limits the substitution to the third occurrence per line; `-n ... p` prints only lines where the substitution happened.

Exact output:

```
2025-06-12 server1: error: disk full; error: retry failed; WARNING: giving up
warning: error error ERROR WARNING
error,error;WARNING - still investigating
error error WARNING error error
```

!! Best wishes !!

Evaluation Scheme

1. Q1 (3):

- Correct meaning of `>` vs `>>` (1)
- Overwrite example command (1)
- Append-to-log example command (1)

If students do not write to create a file when it is not present, 1/2 mark will be deducted.

2. Q2 (3):

- Meaning of `^` (start) and `$` (end) — 0.5 each (1)
- Correct regex `^[0-9]{3}$` (uses one class, anchored) (2)

3. Q3 (3):

- Definitions: `$0`, `$1`, `NF`, `NR` — 0.5 each (2)
- One-liner printing last field: `awk '{print $NF}'` (1)

if students are Line is not written at last for all variable then marks deducted

4. Q4 (3):

- Two modes (Normal, Insert, visual) — 0.5 each (1)
- Two motions (e.g., `w`, `$` or similar) — 0.5 each (1)
- Global substitute command `:%s/foo/bar/g` (1)

5. Q5 (3+2):

- Correct ERE with anchors and line numbers: `grep -nE '^(Error|Warning):' sample.txt` (3)
- Exact four output lines with correct numbers (2)

6. Q6 (3+3+1):

- Correct arithmetic printing sum (3)
- Argument count validation and proper usage message format (3)
- Clear run instructions (chmod / invocation) (1)

If students give space between a variable name and the = sign in shell script, it is not allowed and 3 marks will be deducted.

7. Q7 (3+3):

- `sed -n 's/\<error\>/WARNING/3p'` (or equivalent) — replaces 3rd only and prints on success (3) (only correct command provide full marks or zero marks)
- Exact four substituted output lines as shown (3)